Tu Vu

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Appointments

Virginia Tech Fall 2024 — *present*
Assistant Professor, Computer Science
Research Interests: *large language models & natural language processing*

Google
Faculty Researcher Spring 2025 — *present*
Google DeepMind Research Scientist 2023 — 2024

Education

University of Massachusetts, Amherst 2016 — 2023
M.S/Ph.D. in Computer Science
Advisor: Mohit Iyyer 2018 — 2023
Thesis committee: Mohit Iyyer, Subhransu Maji, Hamed Zamani, Thang Luong, Colin Raffel

Vietnam National University, Hanoi 2009 — 2013
B.S. Honors Program in Computer Science
Highest distinction (class rank: 1st)

Professional Experience

Google DeepMind Fall 2022 — Spring 2023
Research Intern *with* Thang Luong & Quoc Le

Google DeepMind Summer 2022
Research Intern *with* Thibault Sellam & Elizabeth Clark

Google DeepMind Winter 2021 — Spring 2022
Research Intern *with* Noah Constant

Google DeepMind Summer 2021 — Fall 2021
Research Intern *with* Daniel Cer & Noah Constant

Google DeepMind Winter 2020 — Spring 2021
Research Intern *with* Thang Luong & Quoc Le

Google DeepMind Summer 2020
Research Intern *with* Grady Simon & Zi Yang & Nan Hua

Microsoft Research Summer 2019
Research Intern *with* Tong Wang & Tsendsuren Munkhdalai & Adam Trischler

Selected Preprints & Publications

*: equal contribution. For my latest research papers, see my Google Scholar profile.

WikiSkill: Compiling Agent Experience into Persistent Knowledge for Skill Evolution
Liyan Tang, Cyrus Rashtchian, Chun-Sung Ferng, Andrew Tomkins, Da-Cheng Juan, **Tu Vu**
Under review @ ICLR 2027
// VentureBeat feature // DAIR.AI's Top AI Papers of the Week

EvoSkill: Automated Skill Discovery for Multi-Agent Systems
Salaheddin Alzubi, Noah Provenzano, Jaydon Bingham, Weiyuan Chen, and **Tu Vu**
COLM 2026
// Pioneering work on skill evolution

SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models
Thinh Pham, Nguyen Nguyen, Pratibha Zunjare, Weiyuan Chen, Yu-Min Tseng, **Tu Vu**
ICLR 2026
// Our benchmark dataset has been used by Google's Gemini, Qwen, DeepSeek, Kimi, and Perplexity

The Master Key Hypothesis: Unlocking Cross-Model Capability Transfer via Linear Subspace Alignment
Rishab Balasubramanian, Pin-Jie Lin, Rituraj Sharma, Mohit Bansal, **Tu Vu**
NeurIPS 2026 Spotlight (Top 1.3% out of 30K+ submissions)

Foundational Autoraters: Taming Large Language Models for Better Automatic Evaluation
Tu Vu*, Kalpesh Krishna*, Salaheddin Alzubi, Chris Tar, Manaal Faruqui, Yun-Hsuan Sung
EMNLP 2024
// The top-performing generative model on RewardBench as of July 15, 2024, trained only on publicly available data

Gemini: A Family of Highly Capable Multimodal Models
Google Gemini Team: Rohan Anil, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew Dai, Anja Hauth, and others including **Tu Vu**
Technical report 2023
// Google AI Blog

FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation
Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong
ACL 2024 Findings
// ZDNET feature // DAIR.AI's Top ML Papers of the Week // Our dataset and method have inspired or been used for the development of Google's Gemini, Perplexity.AI's Online LLMs, You.com, and Contextual AI's RAG 2.0

The Flan Collection: Designing Data and Methods for Effective Instruction Tuning
Shayne Longpre, Le Hou, **Tu Vu**, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc Le, Barret Zoph, Jason Wei, and Adam Roberts
ICML 2023
// Pioneering work on large-scale instruction tuning // Google Research Blog

Mixture-of-experts meets instruction tuning: A winning combination for large language models
Sheng Shen, Le Hou, Yanqi Zhou, Nan Du, Shayne Longpre, Jason Wei, Hyung Won Chung, Barret

Zoph, William Fedus, Xinyun Chen, **Tu Vu**, Yuexin Wu, Wuyang Chen, Albert Webson, Yunxuan Li, Vincent Zhao, Hongkun Yu, Kurt Keutzer, Trevor Darrell, and Denny Zhou
ICLR 2024
// Pioneering work on combining mixture-of-experts and instruction tuning

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer
Tu Vu, Brian Lester, Noah Constant, Rami Al-Rfou, and Daniel Cer
ACL 2022
// Pioneering work on soft prompt tuning & transfer // Headlines of Google AI's Natural Language Accelerated Newsletter Q1, 2022

Exploring and Predicting Transferability across NLP Tasks
Tu Vu, Tong Wang, Tsendsuren Munkhdalai, Alessandro Sordoni, Adam Trischler, Andrew Mattarella-Micke, Subhransu Maji, and Mohit Iyyer
EMNLP 2020
// Pioneering work on exploring task transferability in the LLM era

Research Awards & Funding

As a lead PI, I have raised $1.2M in research funding from NSF and industry labs

Technology-Enhanced Learning Grant Award (*$4,140)* 2026
PI: Tu Vu

Google Cloud credits (*$20,000)* 2026
PI: Tu Vu

NSF CISE Future Computing Research (Future CoRe) Award (*$821,351)* 2026
PI: Tu Vu, Co-PI: Mohit Iyyer; Personal share: *$421,351*

Amazon - VT Faculty Research Award (*$75,000)* 2026
PI: Tu Vu

Commonwealth Cyber Initiative (CCI) Award (*$35,000)* 2026
PI: Bimal Viswanath, Co-PI: Tu Vu; Personal share: *$5,000*

Gemini Academic Program Award (*$30,000 Gemini credits)* 2026
PI: Tu Vu, Student: Noah Provenzano

MGI Scaling Scholarship Award (*$4,000)* 2026
PI: Tu Vu

Sentient Labs (*$35,000)* 2026
PI: Tu Vu

Google Research Award (*$50,000)* 2025
PI: Tu Vu

Google DeepMind Research Award (*$30,000)* 2025
PI: Tu Vu

Amazon - VT Faculty Research Award (*$130,000)* 2025
PI: Tu Vu

Google AI Studio Tier 1 2025
PI: Tu Vu

VT New Faculty Mentoring Grant (*$1,500*) 2025
PI: Tu Vu

Adobe Research Gift (*$5,000*) 2024
PI: Tu Vu

Advising

PhD advisees:

Mikaela Cankosyan, 1st year PhD student — Fall 2026 — *present*

Noah Provenzano, 1st year PhD student — Fall 2026 — *present*

Weiyuan Chen, 2nd year PhD student — Fall 2025 — *present*

Yu-Min Tseng, 2nd year PhD student — Fall 2025 — *present*

Rishab Balasubramanian, 3rd year PhD student — Fall 2024 — *present*

Thinh Pham, 3rd year PhD student — Fall 2024 — *present*

Quyet Do, 3rd year PhD student — Fall 2024 — *present*

Pin-Jie Lin, 3rd year PhD student — Fall 2024 — *present*

Masters advisee(s):

Rituraj Sharma, 2nd year MS student — Fall 2025 — *present*

Undergrad advisee(s):

Nguyen Nguyen, senior UG student — Spring 2025 — *present*

High school advisee(s):

Samson Ou, Blacksburg High School — Fall 2026 — *present*

Alumni:

Jaydon Bingham, senior UG student — Spring 2026

Christian Calvo, senior UG student — Spring 2026

Others:

Zhenting Qi, Student Researcher @ **Google** — Summer — Fall 2025

Prateek Yadav, Research Intern @ **Google DeepMind** — Summer 2024 — Spring 2025

Simeng Han, Student Researcher @ **Google DeepMind** — Summer 2024 — Spring 2025

Salaheddin Alzubi, Masters student @ **UMass Amherst** — Fall 2022 — Spring 2023

Dheeraj Mekala, PhD student @ **UCSD** — Spring — Summer 2022

On committee:

Parshin Shojaee, VT CS PhD — 2025

Gaurav Srivastava, VT CS MS — 2025

Hani Alomari, VT CS PhD — 2025 — present

Mridul Khurana, VT CS PhD — 2025 — present

Xiaohang Tang, VT CS PhD — 2025 — present

Teaching

CS 5624: Natural Language Processing — Fall 2026

CS 6804: Frontier AI Systems — Spring 2026

CS 4804: Introduction to AI — Fall 2025

CS 5624: Natural Language Processing — Spring 2025

Recent Invited Talks & Lectures

Transformers — September 2026
Invited lectures @ **The New Turing Institute**

Efficient Model Development through Fine-tuning Transfer — October 2025
Lightning talk @ **The Amazon-Virginia Tech AI Workshop**

Attention Mechanisms and Transformers — August 2025
Invited lectures @ **The New Turing Institute**

SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models — August 2025
Featured invited speaker, **Open AGI Symposium @ UC Berkeley**

Multimodal LLMs & Test-time Scaling — June 2025
Invited lecture @ **The New Turing Institute**

Efficient Model Development in the Era of Large Language Models — November 2024
Qualcomm

Efficient Model Development in the Era of Large Language Models — October 2024
Mila / McGill NLP Seminar

Efficient Adaptation of Large Language Models — November 2023
Graph Neural Networks Reading Group, **Google**

Effective and Efficient Transfer Learning in the Era of Large Language Models — Spring 2023
Faculty job talk at multiple institutions

Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation — October 2022
Parameter Efficient Tuning Methods Sync, **Google**

Transfer Learning with Large-scale Language Models — August 2022
Lecture @ **The New Turing Institute**

The Appeal of Parameter-efficient Transfer Learning — June 2022
Natural Language Accelerated Team, **Google**

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer — December 2021
Parameter Efficient Tuning Methods Sync, **Google**

External service

Senior Area Chair for NAACL 2027

Action Editor for TMLR

Area Chair for ICLR 2027; ACL 2026, 2025, 2024; EMNLP 2026, 2025, 2024; NAACL 2025; EACL 2027, 2026; COLING 2025

Session Chair (Machine Learning for NLP) @ EMNLP 2024

Program Committee/Reviewer for Nature 2025, NeurIPS 2025, 2024; ICLR 2026, 2025; ICML 2026, 2025; TL4NLP@NeurIPS 2022; MCDC@ICLR 2025; COLM 2026, 2025, 2024; AAAI 2023; ACL 2022, 2021, 2020, 2019; EMNLP 2022, 2021; NAACL 2022, 2021; COLING 2020; CoNLL 2019; INLG 2020, 2019

Institutional service

CS@VT Faculty Search Committee 2025 — 2026

CS@VT Graduate Admissions Committee 2025 — 2026

Selected Media

VT News: Building AI that speaks your language 2026

WikiSkill: VentureBeat 2026

VT News: Thinking small: How small language models could lessen the AI energy burden 2026

VT News: Amazon-VT Initiative Faculty Award 2025

Gemini: Google AI Blog 2023

FreshLLMs: ZDNET 2023

The Flan Collection: Google Research Blog 2023

SPoT: Headlines of Google AI's Natural Language Accelerated Newsletter Q1, 2022

Other awards & honors

Google Student Researcherships 2020 — 2023

UMass Amherst Graduate Assistantships 2016 — 2023

Honda Y-E-S Award for young engineers and scientists, Vietnam 2013
// in the top 10 nationally

Outstanding Academic and Co-curricular Achievements, Vietnam National University 2013

Prominent Young Figure Award, Vietnam National University 2010 & 2012

First Runner-up Prize, International Programming Contest, Japan 2011
// ranked 2nd among 64 teams internationally

Outstanding Young Talent of the Capital City, Vietnam 2010
// in the top 100 most outstanding young talents selected from a wide range of fields

Champion Prize, National Mathematical Olympiad, Vietnam 2010
// ranked 1st among more than 600 contestants nationally

A number of prizes in National/International Olympiads 2009 — 2013
(in both Mathematics and Informatics)

A number of academic scholarships for undergraduate students 2009 — 2013

Patents

*: original inventor

Frozen Model Adaptation Through Soft Prompt Transfer
Tu Vu*, Daniel Cer, Noah Constant, Brian Lester, Rami Al-Rfou
U.S. Patent Application, 17/863,840

Task Augmentation and Self-training for Improved Few-shot Learning
Thang Luong, **Tu Vu***, Quoc Le, Grady Simon
U.S. Patent Application, 17/826,690